Module Guide for The Crazy Four

Team #25, The Crazy Four

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1 Revision History

Date	Version	Notes
Nov 6	Alvin, Ammar, Ruida, Jim	Initial draft of basic MG (not in report)
Nov 7	Above & Chris Schankula	Review and feedback of basic MG
Nov 8	Alvin	Module Hierarchy and decomposition
Nov 9	Alvin	Updated API's for game action modules
Nov 13	Alvin	modified module decomposition to remove incorrect content
Nov 13	Ammar	added ACs, AC Traceability, Use Hierarchy, UI, Design of CPs; WebSocket, Timeline
Jan 11	Alvin	rev0 - remove m20-22,update m5, hierarchy, add dag graph
Jan 11	Ammar	added UI sketches and protocol section
Jan 21	Alvin	fix dag graph display issue
Jan 21	Ruida	fix protocol section more high level, fix image display issue
Apr 5	Ruida	Rev1: label UCs, reclassify modules, add M20 (Numeral System), fix DAG, expand timeline, update for NumeralSystem/MathChallenge/new actions

2 Reference Material

This section records information for easy reference.

2.1 Abbreviations and Acronyms

symbol	description
AC	Anticipated Change
API	Application Programming Interface
CSS	Cascading Style Sheets
DAG	Directed Acyclic Graph
DOM	Document Object Model
FR	Functional Requirement
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
M	Module
MG	Module Guide
NFR	Non-Functional Requirement
OS	Operating System
R	Requirement
REST	Representational State Transfer
SC	Scientific Computing
SR	Safety Requirement
SRS	Software Requirements Specification
UC	Unlikely Change
UI	User Interface
WS	WebSocket

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3 Introduction

Decomposing a system into modules is a commonly accepted approach to developing software. A module is a work assignment for a programmer or programming team (Parnas et al., 1984). We advocate a decomposition based on the principle of information hiding (Parnas, 1972). This principle supports design for change, because the "secrets" that each module hides represent likely future changes. Design for change is valuable in SC, where modifications are frequent, especially during initial development as the solution space is explored.

Our design follows the rules layed out by Parnas et al. (1984), as follows:

- System details that are likely to change independently should be the secrets of separate modules.
- Each data structure is implemented in only one module.
- Any other program that requires information stored in a module's data structures must obtain it by calling access programs belonging to that module.

After completing the first stage of the design, the Software Requirements Specification (SRS), the Module Guide (MG) is developed (Parnas et al., 1984). The MG specifies the modular structure of the system and is intended to allow both designers and maintainers to easily identify the parts of the software. The potential readers of this document are as follows:

- New project members: This document can be a guide for a new project member to easily understand the overall structure and quickly find the relevant modules they are searching for.
- Maintainers: The hierarchical structure of the module guide improves the maintainers' understanding when they need to make changes to the system. It is important for a maintainer to update the relevant sections of the document after changes have been made.
- Designers: Once the module guide has been written, it can be used to check for consistency, feasibility, and flexibility. Designers can verify the system in various ways, such as consistency among modules, feasibility of the decomposition, and flexibility of the design.

The rest of the document is organized as follows. Section 4 lists the anticipated and unlikely changes of the software requirements. Section 5 summarizes the module decomposition that was constructed according to the likely changes. Section 6 specifies the connections between the software requirements and the modules. Section 7 gives a detailed description of the modules. Section 8 includes two traceability matrices. One checks the completeness of the design against the requirements provided in the SRS. The other shows the relation between anticipated changes and the modules. Section 9 describes the use relation between modules.

4 Anticipated and Unlikely Changes

This section lists the project changes that we expect ("Likely Changes") and those we consider unlikely within the capstone timeframe ("Unlikely Changes"). Where applicable each anticipated change (AC) has been given an identifier to support traceability and to guide module secrets.

4.1 Likely Changes (each becomes an AC / module secret)

AC1 — Rule Engine Changes: The validation and behavioural details of the game's rule engine (e.g., how special cards behave, new house rules or alternate base arithmetic rules).
Rationale: Rules are likely to be refined during playtesting and instructor feedback, so they should be isolated in the M16 module.

AC2 — Numeric Base Support (Dozenal and beyond): Support for additional numeral systems (Octal, Hex, etc.), and how scores and hints convert and present values.
Rationale: We intend the product to be extensible to multiple bases; the conversion logic and scoring changes must be encapsulated in M18, M20, M17, and M13.

AC3 — Scoring Formula / Presentation: Changes to how round or match scoring is calculated, aggregated or displayed (e.g., bonus rules, alternate tallying).
Rationale: Scoring may evolve after playtests; keep scoring logic isolated in M17.

AC4 — UI and Interaction Patterns: Layout, hint presentation, animations, accessibility adaptations, and onboarding flow.
Rationale: UI polish and accessibility tweaks are frequent after usability sessions; the M8, M11, and M12 should hide these secrets.

AC5 — Persistence Details: Changes to the persistent data model (profiles, history, telemetry format) and storage strategy.
Rationale: Data collection needs evolve; the M5 module hides these details.

AC6 — Real-time Messaging Strategy: Room lifecycle, message batching, reconnection rules, and state-resync semantics.
Rationale: Real-time policies affect multiplayer stability; isolate them in M2 and M7.

AC7 — Authentication and Session Policies: Token lifetime, guest-session semantics, password storage scheme, and auth flows.
Rationale: Security/configuration choices can change; these are hidden by M4 and the M1 interface.

4.2 Unlikely Changes

These items are not expected to change during the capstone, so they are *not* treated as module secrets and are documented rather than encapsulated in a separate module secret:

UC1 — Target Runtime Platforms: Modern desktop browsers remain the deployment target for the deliverable.

UC2 — Primary Tech Stack: Web frontend framework, Server-side JavaScript runtime, relational database for persistence.

UC3 — Core Gameplay Pattern: The turn-based matching mechanics (match by suit / match by value / sum-to-target in the active numeral system) are a stable product decision.

UC4 — Non-functional Goals Baseline: Performance targets (sub-300 ms updates), basic security expectations (TLS usage for network), and the expectation of GitHub-based CI.

5 Module Hierarchy

This section provides an overview of the module design. Modules are summarized in a hierarchy decomposed by secrets in Table 1. The modules listed below, which are leaves in the hierarchy tree, are the modules that will actually be implemented.

M1: API Module

- Provides stateless HTTP (REST) endpoints for auth and profile management. Active gameplay is handled separately via the Real-time Gateway.

M2: Real-time Gateway Module

- Manages stateful WebSocket connections for live gameplay, state syncing, in-game chat, and restart negotiation.

M3: Matchmaking Module

- Handles game lobby creation, joining, leaving, and starting a match with a chosen numeral system.

M4: Authentication Module

- Manages user identity, password hashing, and session token generation.

M5: Repository Module

- Abstracts all persistent storage operations for creating, reading, updating, and deleting data, shielding callers from the underlying storage technology.

M6: Audit Module

- Logs important server-side events for debugging and security.

M7: Real-time Client Module

- Establishes and maintains the client-side WebSocket connection; sends/receives game events.

M8: Application Shell Module

- The main component providing global layout, navigation, and state.

M9: Authentication Client Module

- Provides the UI and logic for login/signup forms.

M10: Lobby View Module

- UI component for displaying, creating, and joining game lobbies; lets the host select the numeral system for the game.

M11: Game Board View Module

- UI component that renders the main game interface (hands, deck, discard pile, challenge popups, chat).

M12: Move Controller Module

- Manages user input (e.g., card clicks) and highlights valid moves; handles challenge answer submission.

M13: Scoreboard View Module

- UI component for displaying round/match scores in the active numeral system.

M14: Profile View Module

- UI component for displaying user statistics and game history.

M15: Game Engine Module

- Manages the core game state (deck, hands, scores, challenges) and turn progression; orchestrates round lifecycle.

M16: Rules Module

- Stateless logic to validate moves (e.g., match suit, rank, or target sum), enforce draw limits, handle wildcards and skip cards, and generate arithmetic challenges.

M17: Scoring Module

- Calculates scores at the end of a round based on the loser's remaining hand and the active numeral system.

M18: Base Conversion Module

- Utility to convert numbers between arbitrary numeral bases using a configurable base specification.

M19: Game Actions Module

- Defines types and structure for player actions (play card, draw, pass, answer challenge, declare suit).

M20: Numeral System Module

- Defines the configuration for each supported numeral system (deck composition, face ranks, wildcard symbols, target sums, scoring parameters).

Level 1	**Level 2**	**Level 3 (Leaf Modules)**
Hardware-Hiding Module		(None)
Behaviour-Hiding Module	Core Domain Logic	M15 M16 M17 M18 M19 M20
	Gameplay Interaction	M3 M10 M11 M12 M13 M14
Software Decision Module	Backend (Server)	M1 M2 M4 M5 M6
	Frontend (Client)	M7 M8 M9

Table 1: Module Hierarchy

6 Connection Between Requirements and Design

The design of the system is intended to satisfy the requirements developed in the SRS. In this stage, the system is decomposed into modules. The connection between requirements and modules is listed in the Traceability Matrix in Section 8 (Table 2). This decomposition ensures that each Functional Requirement (FR), Non-functional Requirement (NFR), and

Safety Requirement (SR) has a clear owner in the design, facilitating implementation and verification.

For example, core gameplay logic (FR-1 to FR-5) is satisfied by the M15 and M16 modules, while the user-facing presentation (FR-7, FR-9) is handled by frontend modules like M13 and M11. Security and data persistence requirements (FR-10 to FR-17, SR-3, SR-8) are satisfied by the backend's M4 and M5 modules.

7 Module Decomposition

Modules are decomposed according to the principle of "information hiding" proposed by Parnas et al. (1984). The *Secrets* field in a module decomposition is a brief statement of the design decision hidden by the module. The *Services* field specifies *what* the module will do without documenting *how* to do it. For each module, a suggestion for the implementing software is given under the *Implemented By* title.

Only the leaf modules in the hierarchy have to be implemented.

7.1 Hardware Hiding Modules

There are no hardware hiding modules in this design. The system runs on standard computing hardware and operating systems which are abstracted by the chosen runtime environments (Browser/Node.js).

7.2 Behaviour-Hiding Module

7.2.1 Game Engine Module (M15)

Secrets: Internal representations of the cards, deck, discard pile, players, and the turn-management state machine; orchestration of round lifecycle including challenge resolution and auto-pass logic.

Services: Creates new matches parameterized by a numeral system, enforces turn order, applies validated moves (including challenge answers), and determines when a round or match is finished.

Implemented By: The Crazy Four (Typed JavaScript)

Type of Module: Abstract Data Type

7.2.2 Rules Module (M16)

Secrets: Exact move-validation criteria, including how matching ranks, suits, target sums, wildcards, skip cards, draw limits, free-play grants, and arithmetic challenges are handled.

Services: Confirms whether a proposed move is legal and enumerates valid moves for a player based on the current game situation and the active numeral system.

Implemented By: The Crazy Four (Typed JavaScript)

Type of Module: Abstract Object

7.2.3 Scoring Module (M17)

Secrets: The scoring equation that converts remaining cards into round points, parameterized by the active numeral system's face-card point values.

Services: Produces the score for the round winner based on the loser's remaining hand, returning both a decimal total and a formatted string in the active base.

Implemented By: The Crazy Four (Typed JavaScript)

Type of Module: Abstract Object

7.2.4 Base Conversion Module (M18)

Secrets: The generic algorithm for converting numbers between arbitrary bases, including digit-symbol mapping, alias handling, normalization, and sign handling.

Services: Translates numeric values to and from any configured numeral base for scoring logic and UI presentation.

Implemented By: The Crazy Four (Typed JavaScript)

Type of Module: Abstract Data Type

7.2.5 Game Actions Module (M19)

Secrets: The canonical data structures and serialization format that represent every move a player can make (play, draw, pass, answer challenge).

Services: Defines and validates the action payloads that flow between client and server, keeping both sides in sync on network contracts.

Implemented By: The Crazy Four (Shared Library)

Type of Module: Abstract Object

7.2.6 Numeral System Module (M20)

Secrets: The configuration parameters for each supported numeral system: deck composition (numeric symbols, face ranks), wildcard symbols, target-sum text, target-score text, and face-card point values.

Services: Provides a complete numeral system definition to the game engine, rules, and scoring modules so that the game can operate in decimal, dozenal, or future bases without changes to those modules.

Implemented By: The Crazy Four (Shared Library)

Type of Module: Abstract Object

7.2.7 Matchmaking Module (M3)

Secrets: The lobby data structures, lifecycle policies, and session handoff logic for assembling game tables.

Services: Creates, lists, and manages lobbies so players can host games with a chosen numeral system, join open lobbies, or leave lobbies.

Implemented By: The Crazy Four (Server-side Runtime)

Type of Module: Abstract Object

7.2.8 Lobby View Module (M10)

Secrets: Layout, styling, and interaction patterns for discovering or hosting lobbies, including numeral system selection.

Services: Shows available rooms, lets players create or join sessions, and triggers matchmaking calls via M1 and M7.

Implemented By: The Crazy Four (Client-side Framework)

Type of Module: Abstract Object

7.2.9 Game Board View Module (M11)

Secrets: Visual composition of the board, card animations, challenge popup rendering, chat display, and responsive behavior across devices.

Services: Draws the playable surface, displays player hands and discard piles, shows active challenges, highlights valid actions per the rule engine, and provides in-game chat.

Implemented By: The Crazy Four (Client-side Framework)

Type of Module: Abstract Object

7.2.10 Move Controller Module (M12)

Secrets: Gesture-handling patterns and UX rules for how players select cards, declare suits, select operations for challenges, or submit challenge answers.

Services: Interprets user intent on the board, performs light validation, and forwards structured actions through the M7.

Implemented By: The Crazy Four (Client-side Framework)

Type of Module: Abstract Object

7.2.11 Scoreboard View Module (M13)

Secrets: Presentation choices for score displays in the active numeral system and animations for round summaries.

Services: Shows standings after each round, presenting scores in the active base format in a clear, accessible format.

Implemented By: The Crazy Four (Client-side Framework)

Type of Module: Abstract Object

7.2.12 Profile View Module (M14)

Secrets: Layout decisions for profile cards, statistics summaries, and sensitive account actions.

Services: Displays player history, stats, and account-management controls, including export or deletion requests tied to FR-15..17.

Implemented By: The Crazy Four (Client-side Framework)

Type of Module: Abstract Object

7.3 Software Decision Module — Backend

7.3.1 API Module (M1)

Secrets: REST endpoint structure, payload schemas, and HTTP conventions for stateless backend capabilities.

Services: Exposes stateless HTTP routes for authentication, profile management, lobby operations, and bootstrapping new games as defined in the SRS. Note: Real-time gameplay is handled by M2.

Implemented By: The Crazy Four (Server-side Runtime)

Type of Module: Abstract Object

7.3.2 Real-time Gateway Module (M2)

Secrets: Real-time messaging strategy, room management, conflict-resolution logic for server-authoritative play, restart negotiation protocol, and in-game chat relay.

Services: Hosts WebSocket connections, validates incoming moves, pushes synchronized game state updates to every participant, relays chat messages, and manages restart/leave flows.

Implemented By: The Crazy Four (Server-side Runtime + WebSocket Lib)

Type of Module: Abstract Object

7.3.3 Authentication Module (M4)

Secrets: Password hashing configuration, credential storage details, and token-signing keys.

Services: Creates accounts, validates logins, manages guest sessions, and issues/verifies tokens used by the rest of the backend.

Implemented By: The Crazy Four (Server-side Runtime)

Type of Module: Abstract Object

7.3.4 Repository Module (M5)

Secrets: The underlying storage technology and the data model design, optimized queries, and data access strategies.

Services: Offers a clean persistence interface for storing players, credentials, match history, and statistics while shielding callers from storage specifics.

Implemented By: The Crazy Four (Server-side Runtime)

Type of Module: Abstract Object

7.3.5 Audit Module (M6)

Secrets: The exact event schema, retention policy, and storage targets for operational logs.

Services: Captures authentication, gameplay, and system events to support debugging, compliance, and user inquiries.

Implemented By: The Crazy Four (Server-side Runtime)

Type of Module: Library

7.4 Software Decision Module — Frontend

7.4.1 Real-time Client Module (M7)

Secrets: Connection lifecycle logic, buffering strategy, and reconnection heuristics for the browser client.

Services: Establishes WebSocket links to the M2, relays user actions, and applies server updates to the local UI state.

Implemented By: The Crazy Four (Client-side Framework)

Type of Module: Abstract Object

7.4.2 Application Shell Module (M8)

Secrets: App-wide navigation plan, shared layout primitives, and global state wiring (theme, auth awareness).

Services: Hosts the consistent chrome of the site and orchestrates routing between major views.

Implemented By: The Crazy Four (Client-side Framework)

Type of Module: Abstract Object

7.4.3 Authentication Client Module (M9)

Secrets: Decisions about secure token storage and the flows for refreshing or clearing credentials in the browser.

Services: Presents login, signup, and logout experiences while coordinating with the M1 for authentication calls.

Implemented By: The Crazy Four (Client-side Framework)

Type of Module: Abstract Object

8 Traceability Matrix

This section shows two traceability matrices: between the modules and the requirements and between the modules and the anticipated changes.

Table 2: Trace Between Requirements and Modules (TblRT)

Requirement (FR/NFR/SR)	**Primary Modules**
FR-1 Start new game	M1, M3, M15, M5, M20
FR-2 Turn management	M15, M16, M19, M2, M7, M12, M11
FR-3 Rule validation	M16, M15, M19, M12, M11
FR-4 Special cards	M16, M15, M19, M12, M11
FR-5 End of game	M15, M17, M13, M5
FR-6 Calculate score	M17, M18, M20, M19, M13
FR-7 Display score	M13, M18, M20
FR-9 Highlight valid moves	M12, M11, M16
FR-10 Account creation	M1, M4, M5, M9
FR-11 Login or Logout	M1, M4, M5, M9
FR-12 Guest mode	M1, M4, M9
FR-13 Credential validation	M4, M1, M5
FR-14 Data storage	M5, M1, M6
FR-15 Data retrieval	M5, M1, M14
FR-16 Data update	M5, M1, M14
FR-17 Data deletion	M5, M1, M14
NFR (Performance)	M2, M15, M7, M11
NFR (Usability)	M11, M8
NFR (Robustness)	M7, M2, M5
NFR (Maintainability)	M16, M17, M1, M6, M20
SR-1 (Dozenal validation)	M16, M18, M15, M20
SR-2 (UI feedback)	M11, M8
SR-3 (Data persistence)	M5, M1
SR-4 (Accurate scoring)	M17, M18, M20
SR-5 (Session recovery)	M7, M2, M4
SR-7 (Encrypted transmit)	M1, M2
SR-8 (Secure storage)	M5, M4
SR-10 (Input validation)	M1, M2, M12

Table 3: Trace Between Anticipated Changes and Modules (TblACT)

Anticipate Change	**Modules**
AC1 Rule Engine Changes	M16, M15
AC2 Numeric Base Support	M18, M20, M17, M16, M13
AC3 Scoring Formula / Presentation	M17, M13
AC4 UI / Interaction Patterns	M8, M11, M12, M13
AC5 Persistence Details	M5, M1, M14
AC6 Real-time Messaging	M2, M7, M15
AC7 Authentication Policies	M4, M9, M1

Notes: Each AC above corresponds to a "secret" that should be implemented and documented within the named module(s). When an AC is realized, the owning module's MIS (or submodule docs) should be updated so that the secret is well described and the code-to-docs trace remains intact.

9 Use Hierarchy Between Modules (Uses Relation)

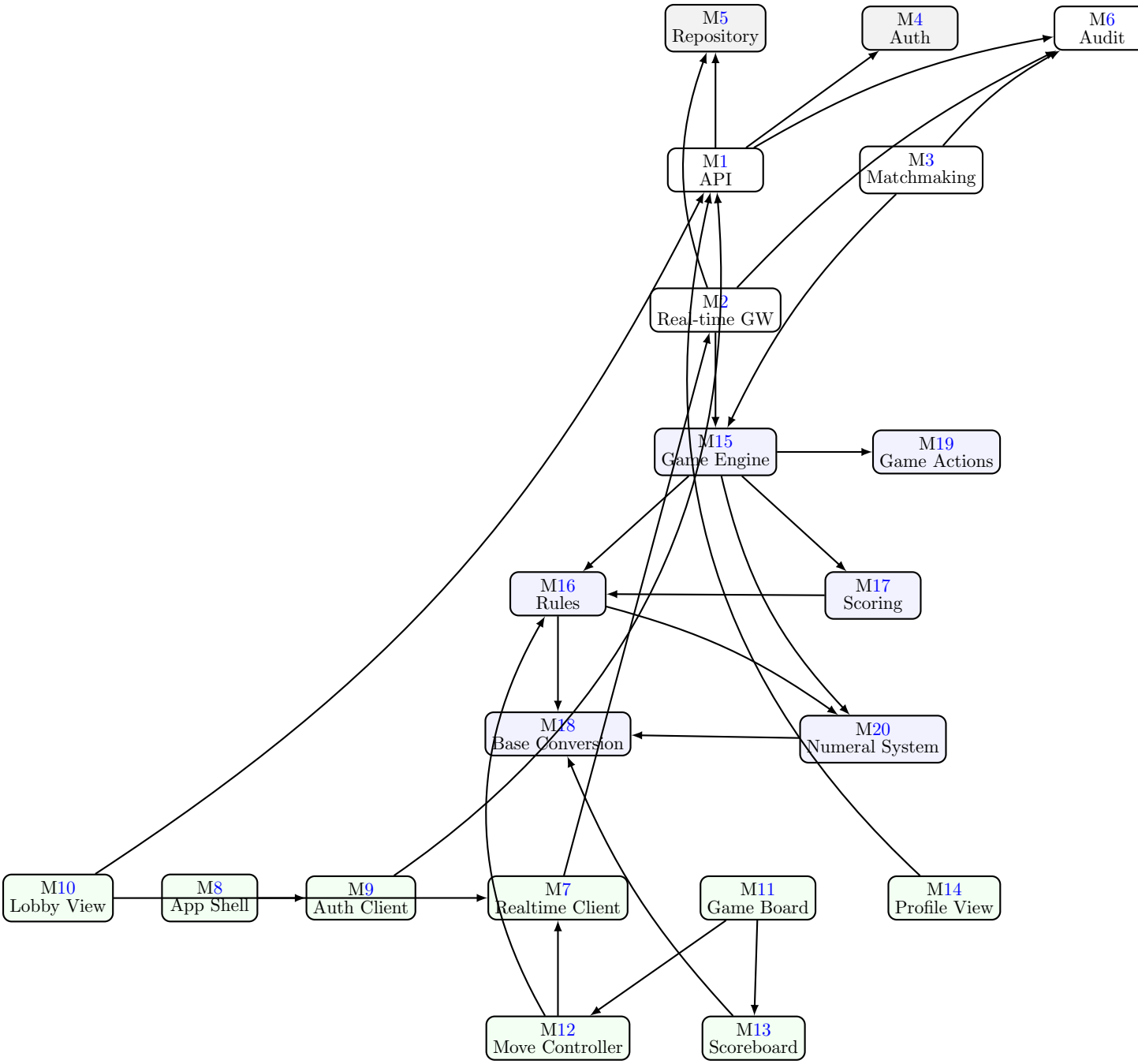

Figure 1: Use hierarchy (DAG) showing primary "uses" relations between modules.

Design rationale: The DAG above shows the server-authoritative pattern: Realtime gateway owns live interactions and forwards validated moves to the Game Engine; the Game

Engine uses the Rules and Scoring modules, and persistence is abstracted via the Repository. The Numeral System module configures the Rules, Scoring, and Base Conversion modules so that the game can operate in any supported base. Frontend views rely on the Realtime client for live updates and the API for non-realtime operations (profile, auth, lobby listing).

10 User Interfaces (UI) — sketches and wireframes

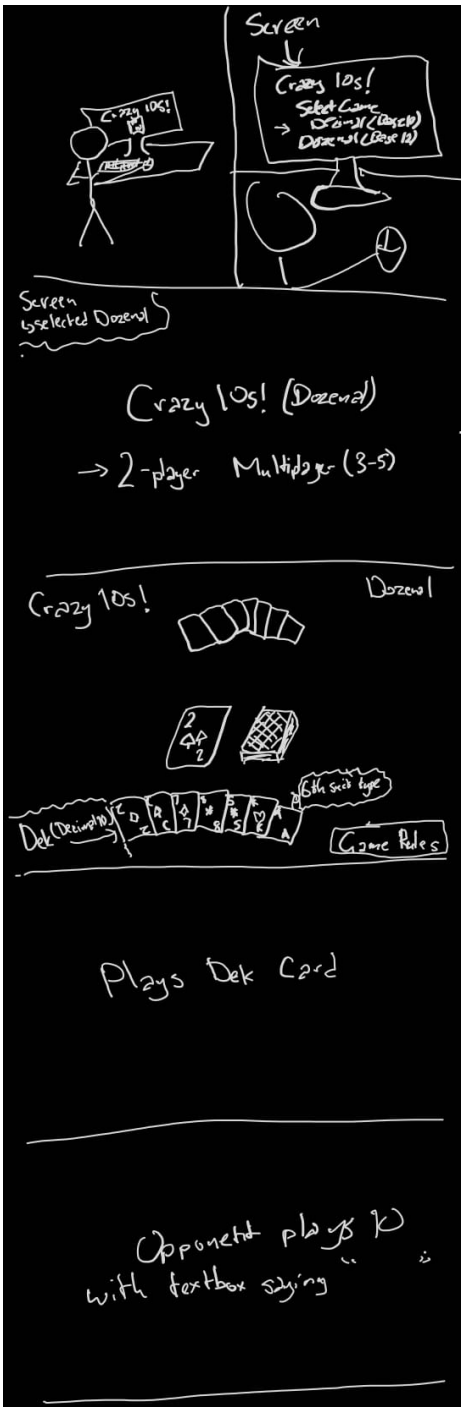

Figure 2: Hand-drawn / mockup: Game board layout.

11 Design of Communication Protocols

The system uses two communication channels between the client (frontend) and the server (backend):

- **Request–response channel (HTTP/REST).** This channel is used for operations that are naturally stateless and non real-time, such as account-related actions (e.g., login/logout), basic metadata queries, and health checks. Using a request–response protocol keeps these operations simple to debug, easy to cache (when applicable), and independent from long-lived real-time connections.
- **Real-time channel (bidirectional, event-driven).** This channel is used for time-sensitive game interactions that require low-latency updates, such as joining/leaving an active match, submitting player actions during gameplay, receiving game state updates, in-game chat, and restart/rematch negotiation. A persistent bidirectional channel avoids polling and supports immediate state synchronization among participants.

Authority and consistency. The server maintains the authoritative game state. Clients do not directly modify the canonical state; instead, clients send *action intents* to the server. The server validates each intent against the game rules and either applies it (followed by pushing the updated state) or rejects it (followed by an error response and/or a state resynchronization). This server-authoritative design prevents rule inconsistencies between clients and reduces the risk of cheating.

Client-side guidance (non-authoritative). To improve usability, the client may compute advisory hints (e.g., highlighting playable cards) based on the most recent state it has received. These hints do not grant permission to act; the server remains the final arbiter of legality. If advisory hints diverge from the server decision, the client UI must defer to the server response and the latest synchronized state.

Session binding and access control. A user session (or equivalent identity) is established through the request–response channel and bound to the real-time connection when needed. The real-time channel only accepts game participation and action submission from authenticated/authorized clients according to the system's access control policy.

Failure handling and recovery. The protocol design supports transient network failures. Upon disconnection and reconnection, the client can re-establish its session and request the latest game state to restore consistency. Errors are surfaced as structured failures (e.g., invalid request, unauthorized access, invalid action, or missing game session) without exposing internal server details.

Applicability. No special hardware–software communication protocol is required for this project; the communication protocols described above cover client–server interaction.

12 Timeline

The development timeline is organized by module, with a target completion date and the responsible team member for each.

Table 4: Detailed Module Implementation Timeline

Module	Description	Target Date	Responsible
M18	Base Conversion	Jan 21, 2026	Ruida
M20	Numeral System definitions	Feb 12, 2026	Jim
M16	Rules (move validation, challenges)	Jan 21, 2026	Ruida
M17	Scoring	Jan 21, 2026	Ruida
M19	Game Actions types	Jan 21, 2026	Ruida,Jim
M15	Game Engine (state machine)	Jan 21, 2026, 2026	Ruida,Alvin
M4	Authentication service	Jan 21, 2026	Ruida
M5	Repository (persistence)	Jan 27, 2026	Ruida
M6	Audit logging	Jan 21, 2026	Ruida
M14	Profile View UI	Jan 27, 2026	Ruida
M1	API routes (REST)	Jan 21, 2026	Ruida
M2	Real-time Gateway (WS)	Jan 21, 2026	Ruida,Alvin
M3	Matchmaking service	Jan 21, 2026	Ruida,Alvin
M12	Move Controller (client)	Jan 21, 2026	Ruida
M7	Real-time Client (WS)	Jan 27, 2026	Ruida
M8	Application Shell	Jan 21, 2026	Ruida
M9	Authentication Client UI	Jan 27, 2026	Ruida,Jim
M10	Lobby View UI	Jan 29, 2026	Ruida,Alvin
M11	Game Board View	Jan 29, 2026	Ruida,Alvin
M13	Scoreboard View	Jan 29, 2026	Ruida,Alvin

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